Shivam Kumar Verma

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Education

Indian Institute of Information Technology, Nagpur — **November 2022 – May 2026**
B.Tech in Computer Science and Engineering — *Nagpur, Maharashtra*

Experience

poiro.com — **July 2026 – Present**
Fullstack Developer — *TypeScript, Python, FastAPI, Next.js, PostgreSQL, Cloudflare R2*

- **Creative Asset Manager Backend:** Built the data model, services, and APIs for the asset layer used by AI-generation surfaces; fixed production endpoint failures and reference-integrity defects that could hide nested folders and AI-lineage assets, and replaced depth-dependent tree traversal with a **single aggregation query**.
- **Storage Integrity and Deduplication:** Engineered storage-verified **SHA-256** asset deduplication with a database uniqueness guarantee and client-side preflight, preventing duplicate R2 writes without trusting client-supplied hashes.
- **Repository Reliability:** Cleared approximately **1,041 Ruff findings across 7 Python trees**, made lint/format checks blocking in CI and pre-push, and added dynamic package discovery so new packages are covered automatically.

klimb.io — **November 2025 – March 2026**
Full Stack Developer Intern — *TypeScript, Node.js, Express.js, PostgreSQL, Redis, AWS*

- **AI Usage Metering:** Designed a PostgreSQL-backed usage-metering and credit ledger across **6+ AI features**, giving product and finance auditable per-feature usage for consumption-based access control.
- **Data Access Optimization:** Rewrote slow queries and restructured recruiter-dashboard data access, reducing p95 latency and memory overhead for candidate-evaluation and interview operations.
- **AI Interview Platform Backend:** Built backend services for a **3+ stage** AI interview platform with automated proctoring and strict candidate-level data isolation across concurrent live sessions.

Projects

Yuvro | GitHub | Demo | *Python, FastAPI, PostgreSQL, Kubernetes, AWS S3, WebSockets*

- **Isolated Project Runtime:** Built Kubernetes-managed runner environments per project, letting users run code, terminals, previews, and databases without dependency conflicts between workspaces.
- **Indexed Workspace Metadata:** Designed an indexed PostgreSQL file-tree and metadata system for users, workspaces, projects, and folders, enabling fast lookups without storing file contents in the database.
- **Content-Addressed Storage:** Implemented **SHA-256 content-addressed S3 storage** with template manifests and garbage collection, avoiding duplicate file storage and repeated hashing for template-created projects.

CodeSM | GitHub | Live | *Node.js, Express.js, BullMQ, Redis, PostgreSQL, Docker, AWS EC2*

- **Asynchronous Evaluation Pipeline:** Built a BullMQ + Redis execution pipeline that persists submissions before background grading, keeping the API responsive during compile and test workloads, with retries for failed jobs.
- **Sandboxed Multi-Language Runtime:** Isolated untrusted C, C++, Java, Go, Python, and JavaScript code in Docker with CPU, memory, process, network, filesystem, timeout, and output limits.
- **Lazy Testcase Processing:** Fetched S3-backed test data one case at a time with early exit on first failure, bounding per-job memory and avoiding unnecessary downloads.

Technical Skills

Infrastructure & Runtimes: Docker, Kubernetes, sandboxed execution, BullMQ, Redis, AWS (EC2, S3), Linux
Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, WebSockets, distributed job processing
Data Systems: PostgreSQL, Redis, MongoDB, Prisma, Cloudflare R2, content-addressed storage, deduplication pipelines
Languages & Automation: TypeScript, JavaScript, Python, C++, Playwright, Postman, Ruff, ESLint, Turbo

Achievements

- **Competitive Programming: Top 5% globally** on LeetCode (**1800+ rating**, 800+ problems solved) · **3-star** on CodeChef.
- **Tantrafiesta 2024: 3rd rank** at IIIT Nagpur's hackathon.